MIL-R-24037/6C(EC)
30 March 1972
SUPERSEDING 1/
MIL-R-24037/6B(EC)
11 February 1970

MILITARY SPECIFICATION

RADIACMETERS, QUARTZ-FIBER SELF-INDICATING RADIACMETER - IM-181( )/PD

1. SCOPE

1.1 This specification covers the detail requirements for an ion-chamber quartz-fiber dosimeter (radiacmeter) capable of measuring and indicating doses of gamma or X-radiation over the range of 0 to 1000 milliroentgens (mR).

2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on date of invitation for bids or request for proposal, forms a part of the specification to the extent specified herein:

SPECIFICATION
MILITARY
MIL-R-24037 - Radiacmeter, Quartz-Fiber Self-Indicating, General Specification.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 The equipment shall be in accordance with MIL-F-24037 and as specified herein.

1/ CHANGES FROM PREVIOUS ISSUE. THE MARGINS OF THIS DOCUMENT ARE MARKED WITH A "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE WERE MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

FSC 6665

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3.2 Scale. The scale shall have 20 equal divisions, each division representing 50 mR. There shall be 11 major marking lines: every other major line above the base line, starting at the first major line shall be marked 0, 200, 400, 600, 800, and 1000 respectively. Every other major line below the base line shall be marked 100, 300, 500, 700 and 1000 respectively. The major lines shall be appropriately marked with arabic numerals. There shall be a minor unnumbered line, shorter than the major lines, midway between each major line. The base line shall be an equally heavy line (same as a major line) perpendicular to and bisecting all scale divisions. The word MILLIROENTGENS shall appear above the scale. The nomenclature of the radiacmeter shall appear above the word MILLI-ROENTGENS. No other markings are permitted.

3.3 Range. The range of the radiacmeter shall be 0 to 1000 mR.

3.4 Leakage. The electrical resistance between the charging electrode and the barrel shall be such that leakage in the absence of radiation shall be not more than 20 mR per day for any relative humidity to 100 percent, and at any temperature between minus 65° and plus 131°F. Leakage shall be taken over a period of 5 days. The initial reading shall be taken 8 to 24 hours after charging. The final reading shall be taken between 108 and 120 hours after initial reading.

3.5 Leakage after heavy irradiation. After a continuous or intermittent exposure totaling 200 R given at dose rates up to 1000 R per hour, electrical leakage in a 4-hour period starting not more than 10 minutes after exposure is completed, shall not exceed 5 percent of full scale. After a 48-hour period, the leakage requirement specified in 3.4 shall apply.

3.6 Accuracy. The radiacmeter shall measure true gamma dose from a cobalt 60, radium, or cesium 137 source within ±10 percent or ±50 mR, whichever is larger. This accuracy requirement shall apply for any source which has been (a) calibrated directly by the National Bureau of Standards, (b) calibrated by the contractor using Victoreen R meters, or equivalent, approved by the command or agency concerned, which have in turn been calibrated by the National Bureau of Standards, or (c) calibrated by a Government activity using Victoreen R meters, or equivalent approved by the command or agency concerned, which have been calibrated by the National Bureau of Standards. Specifically included in this spread are the usual imprecisions involved in correlating source strengths with the standard National Bureau of Standards sources.

3.6.1 Precision. Each radiacmeter shall be capable of reproducing its own readings to 5 percent when subjected to a series of calibration runs at midscale.

3.7 Energy dependence. Except at the energies represented by cesium 137 and cobalt 60, where the requirements of 3.6 shall apply, the radiacmeter shall indicate true gamma dose within ±15 percent or 50 mR,

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whichever is larger, for X- or gamma radiation of any energy from 80 kiloelectrovolts (keV) to the energy of cobalt 60 gammas. Over the range 80 to 50 keV, the radiacmeter response shall decrease smoothly; at 50 keV and at all X- and gamma radiation energies below 50 keV the response shall be not greater than 50 percent of true dose. As used here, 80 and 50 keV refer to effective energies of X-ray beams from a constant voltage X-ray machine, with appropriate tube voltage and filtration similar to National Bureau of Standards heavily filtered X-ray techniques.

3.8 Ion chamber saturation and rate dependence. The radiacmeters, when subjected to dose rates up to 5000 R per hour, shall not vary from their normal calibration accuracy by more than ±50 mR.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be as specified in MIL-R-24037, and as specified herein.

4.2 Leakage. The radiacmeter shall be fully charged and tested for a period of 5 days at an ambient temperature of at least 100°F. The deflection of the electroscope fiber shall not exceed 100 mR during this test. The radiacmeters may be stored under charge for 8 to 24 hours prior to this test if desired. The fiber shall be charged positively in this and all other tests.

4.3 Calibration at midpoint only. A source, calibrated in accordance with 3.6 shall be used. Temperature may be ambient room temperature. A dose of 500 mR shall be delivered to the radiacmeter. The reading shall be not greater than 550 mR nor less than 450 mR.

4.3.1 Precision. The instrument shall be tested for compliance with 3.6.1. The radiacmeter shall be identically exposed for five exposures. All readings shall fall within 5 percent of the average reading for the particular instrument.

4.4 Calibration over the entire range. The instrument shall be tested as specified in 4.3 at 4 points: 200, 400, 600 and 800 mR. Readings shall be as specified in 3.6.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in MIL-R-24037.

6. NOTES

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6.1 The notes specified in MIL-R-24037 are applicable to this specification.

Preparing activity:
Navy - EC

(Project 6665-N297)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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